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8 July 1977
SUPERSEDING
MIL-R-45843A
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MILITARY SPECIFICATION

RIMS, CURING, TIRE MOLD VULCANIZER

This specification is approved for use by the US Army Armament Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers curing rims utilized for supporting the curing tubes during tire vulcanizing operations.

1.2 Classification. The curing rims covered by this specification shall be of the following types, as specified (see 6.2.1).

Type I - Road rim
Type II - Collapsible inside curing rim

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-14625 - Curing Bags and Tubes, Tire Repair, Recapping, and Retreading

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Materiel Readiness Command, ATTN: DRSAR-LEE-S, Rock Island, IL 61201 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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MIL-V-45554 - Vulcanizing Equipment, Including Related Items, Tire and Tube, Rebuild and Repair; Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

NBS Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2.1), the supplier shall furnish one complete curing rim for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The curing rims shall be new and of the latest design meeting the requirements of this specification. All parts subject to wear or replacement shall be accessible.

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3.2.1 Type I. The curing rim shall be road rim- or outside-type curing rim designed for mounting on the bead of the tire around the outside of the tire casing.

3.2.2 Type II. The curing rim shall be of a collapsible unit designed for inserting and mounting inside the tire casing.

* 3.2.3 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the curing rims presenting safety hazards which can be guarded without substantial interference to operation of the curing rims. All guards shall be removable to provide easy access of guardec parts. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1).

* 3.3 Construction. The curing rims shall be complete so that when installed they can be used for the operations specified herein.

* 3.4 Parts and materials. All parts and materials used in fabrication shall be new and free of defects and imperfections that might affect the serviceability or appearance of the finished product. Contact between dissimilar metals, especially aluminum or magnesium in contact with steel, copper, or brass, shall be avoided wherever possible.

* 3.4.1 Interchangeability. All parts shall be manufactured to such standards as will permit replacement and adjustment without modification of parts or machine.

* 3.4.2 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No process shall be used for reclaiming any defective part.

* 3.4.3 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. These operations shall not be employed as a repair measure for defective parts.

* 3.4.4 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

* 3.4.5 Threads. All threads shall be in the inch system and shall comply with NBS Handbook H28.

* 3.4.6 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with means for lubrication.

3.5 Performance and product characteristics. The curing rims shall embody the following performance and product characteristics.

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3.5.1 Type I. The curing rim shall consist of two sections which separate between the flanges to facilitate the mounting and demounting operations. The flanges shall securely fit against the tire for retaining the curing tubes in position during the curing operation of tire repairs. The sections shall be installed on the tire without requiring power tools and shall be capable of withstanding the curing operational loads imposed. The curing rim shall incorporate facilities for adjusting the width of the flanges. The curing rim shall be made of aluminum alloy or steel. When installed, the curing rim shall permit the inflating and deflating of the curing tube. The curing rim shall provide distortion-free cures and shall not damage the curing tube and tire. The curing rim shall be capable of being utilized with the type II, class 1 curing tubes of MIL-C-14625.

3.5.2 Type II. The curing rim shall be composed of assembled collapsible or folding segments to permit installing the rim inside the tire casing against the curing tube. The rim segments shall be capable of being secured (locked) in position after positioning inside the tire to provide a rigid curing rim to withstand the curing operation loads imposed without loosening during the tire repair curing operation. The rim segments shall be secured in position without the use of power tools. The curing rim segments shall be made of aluminum alloy or steel. When installed, the curing rim shall permit the inflating and deflating of the curing tube. The curing rim shall provide distortion-free cures and shall not damage the tire and curing tube. The curing rim shall be capable of being utilized with the type II, class 2 curing tubes of MIL-C-14625.

3.5.3 Capacity. The curing rims shall be for use with tire size (diameter and width) as specified by the procuring activity (see 6.2.1).

3.5.4 Excessive pressure requirements. The curing rims shall be capable of withstanding without damage the following hydrostatic pressures for a period of 5 minutes; 250 pounds per square inch gage (psig) for curing rims for tire sizes of 7.00-16 and less and 300 psig for curing rims for tire sizes larger than 7.00-16.

3.6 Special tools. The curing rims shall be furnished with the required special tools to perform the operations specified herein. The quantity of special tools shall be specified by the procuring activity (see 6.2.1).

* 3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing of the curing rims shall be in accordance with the supplier's standard practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall have been cleaned to be free of all foreign matter detrimental to painting, and at least one coat of primer and one coat of enamel have been applied.

* 3.8 Identification marking. The curing rims shall be marked for identification in accordance with MIL-STD-130, and unless otherwise specified (see 6.2.1), shall include the National Stock Number.

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* 3.9 Technical data. Data shall be furnished as specified (see 6.2.2).

* 3.10 Workmanship. Standards of workmanship shall be such as to assure equipment of a neat appearance, having the stability and efficient operating characteristics as found in commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

* 4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed in accordance with the test conditions specified in applicable test method document or applicable paragraph(s) of the contract.

* 4.4 First article inspection. First article inspection shall be applied to the preproduction model or initial production item noted (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination of one complete curing rim for defects described in classification of defects (4.5.2) and all tests under 4.6. Failure of the first article to pass the inspection shall be cause for rejection.

* 4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examination specified in 4.5.2, the tests specified in 4.5.3, and the inspection specified in 4.7.

4.5.1 Quality conformance inspection sampling.

* 4.5.1.1 Acceptance lot. An acceptance lot shall consist of all curing rims of one type from one manufacturer submitted for acceptance at one time.

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* 4.5.1.2 Inspection method. Unless otherwise specified, all inspections shall be performed in accordance with MIL-STD-105, Inspection Level II.

* 4.5.2 Examination. The curing rims shall be examined in accordance with 4.5.1, using classification of defects and acceptable quality level (AQL) as follows:

Classification of defects

Characteristics	Requirement Paragraph	AQL Percent Defective Major	Minor
101 Safety devices	3.2.3	2.5	
102 Castings or forgings	3.4.2	2.5	
103 Welding, brazing, and soldering	3.4.3	2.5	
104 Fastening devices	3.4.4	2.5	
105 Curing rim details			
Type I	3.5.1	2.5	
Type II	3.5.2	2.5	
106 Capacity	3.5.3	2.5	
107 Workmanship	3.10	2.5	
108 Construction	3.3	2.5	
109 Parts & Materials	3.4	2.5	
110 Interchangeability	3.4.1	2.5	
201 Threads	3.4.5		4.0
202 Protective Finish	3.7		4.0
203 Markings	3.8		4.0
204 Lubrication	3.4.6		4.0

* 4.5.3 Quality conformance test. From each lot which has passed the examination specified in 4.5.2, samples for acceptance test shall be selected in accordance with inspection level S-3 of MIL-STD-105. The samples selected shall be subjected to the tests (a) and (b) below. Failure to pass either test shall be cause for rejection of the lot.

(a) Functional test (see 4.6.1).
(b) Excessive pressure test (see 4.6.3).

* 4.6 Test methods.

* 4.6.1 Functional test. Utilizing the curing rim with the appropriate tire and curing tube, the curing rim shall be positioned in accordance with the manufacturer's instructions and the curing tube shall be inflated until fully rounded out. The curing rim shall be observed to determine compliance with the mounting and demounting requirements specified in 3.5.1 or 3.5.2.

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* 4.6.2 Operational test. In accordance with the manufacturer's instructions, the curing rim shall be utilized with the appropriate built-up tire (prepared for curing), applicable curing tube, and vulcanizing mold for performing the tire curing operation. On completion of the curing operation, the curing tube and tire shall be examined to determine compliance with 3.5.1 or 3.5.2.

* 4.6.3 Excessive pressure test. Utilizing the curing rim with the applicable tire size and curing tube, and securing the tire between plates or by other adequate holding means, the curing rims for tires of 7.00-16 size and less shall be subjected to a hydrostatic pressure of 245 to 250 psig, and the curing rims for tires larger than 7.00-16 size shall be subjected to a hydrostatic pressure of 295 to 300 psig. The pressure shall be applied for a period of not less than 5 minutes and on conclusion of the test, the curing rim shall be examined for damage to determine compliance with 3.5.4.

* 4.7 Packaging inspection. The sampling and inspection of the preservation-packaging, packing, and container marking shall be in accordance with the requirements of MIL-V-45554.

5. PACKAGING

5.1 Packaging (preservation, packing, and marking). Level A or B, or commercial packaging shall be in accordance with MIL-V-45554, as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The curing rims covered by this specification are intended for use in base shops for supporting curing tubes during tire vulcanizing operations.

* 6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

* 6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) First article, when required (see 3.1).
(d) Specify exceptions and additional safety and health requirements (see 3.2.3).
(e) Curing rim size required (see 3.5.3).
(f) Special tool(s), quantity required (see 3.6).
(g) Painting, if different (see 3.7).
(h) Identification marking, if different (see 3.8).
(i) Inspection conditions, if different (see 4.3).

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(j) First article inspection, if different (see 4.4).
(k) Selection of applicable level of protection (A or B), or commercial, as specified (see 5.1).

* 6.2.2 Data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.9).

* 6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-45843B - Rims, Curing, Tire Mold Vulcanizer

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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